

Perspectives on Accelerated Computing and Machine Learning for Engineering Simulation

GPUs, surrogates, lessons from large language models, and agentic engineering

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Four claims for the next 35 minutes

1. Accelerated computing

High-fidelity simulation has become affordable, but the gain depends on the algorithm as much as the hardware. [Session 1]

2. Surrogate models

Learned models are now accurate enough to be useful, and only as good as their training data. [Session 2]

3. Lessons from large language models

Pre-training, post-training and scaling laws say what a simulation foundation model would cost and need.

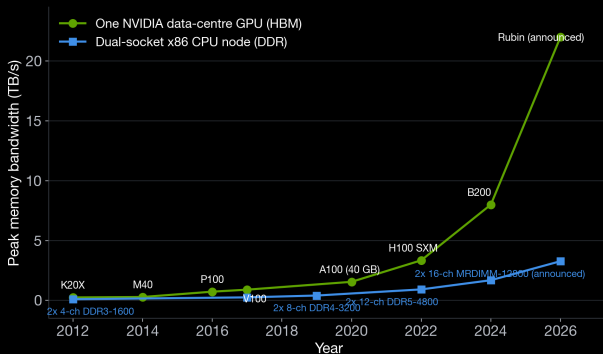
4. Agentic engineering

Agents already run and port CFD workflows; their value is bounded by the speed and trust of the tools they call. [Sessions 3 to 5]

Examples chosen from wind, wave and marine engineering wherever the published evidence allows.

Memory bandwidth is the quantity that matters for most solvers

- Finite-volume, SPH and lattice Boltzmann kernels move far more bytes than they compute. They are bandwidth bound.
- Since 2012 a dual-socket x86 node has gained about 30x, mostly from more channels and multiplexed DIMMs; a single data-centre GPU about 90x.
- Consequence: an 8-GPU node has 30 to 40 times the bandwidth of a dual-socket CPU node, so it can replace tens of CPU nodes for a bandwidth-bound algorithm.

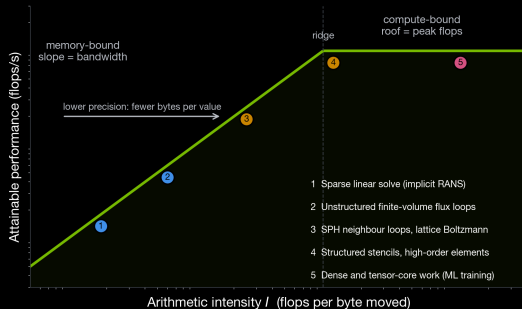


Peak theoretical bandwidth from vendor specifications. CPU node: two sockets, all channels populated at the top supported data rate. 2026 points

are announced products. Sustained bandwidth is lower on both.

Roofline: the algorithm decides how much of the hardware you get

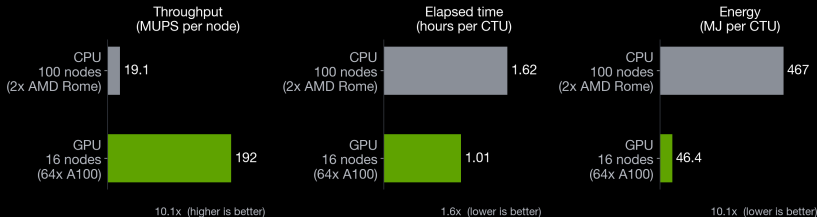
- Arithmetic intensity I (flops per byte) places every kernel on the roofline.
- Explicit, structured, matrix-free and particle methods sit further right and scale well.
- Implicit unstructured solvers live in sparse linear algebra at low I ; they gain from bandwidth alone.
- Lower precision moves a kernel to the right at no algorithmic cost. It is not free in accuracy; next slides.



Today's session shows both routes: [O1 uDALES] [O2 HOS-SPH] [O3 DualSPHysics on Kokkos] [O4 CHAPSim] [O5 gpuFOAM]. Xavier Trias goes deeper on this in [Keynote 3, tomorrow].

Measured evidence: the same 325M-cell high-lift WMLES on CPU and GPU

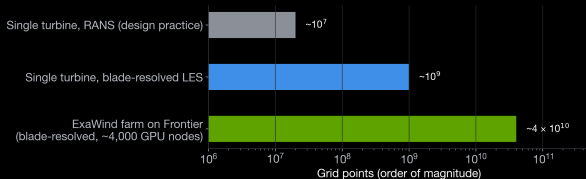
Same 325M-cell high-lift WMLES case, FP64, NASA LAVA solver



- Three different measures. Throughput per node and energy improve by an order of magnitude; elapsed time by less because the CPU allocation was six times larger.
- Energy per convective time unit is the number that will matter most to UK facilities.

NASA LAVA, HLPW-5 Case 2.4, FP64. AIAA 2025-0887, Table 3. CPU: 100 nodes of 2x AMD Rome. GPU: 16 nodes of 4x NVIDIA A100.

Wind energy at exascale: what GPUs made possible



ExaWind on Frontier (NREL, ORNL, Sandia)

- Nalu-Wind (near-body, unstructured) coupled to AMR-Wind (off-body, block-structured) through overset grids.
- Blade-resolved simulation of a multi-turbine farm using around 4,000 of Frontier's 9,408 GPU nodes and tens of billions of grid points.
- Open data: 16-turbine and 4-turbine blade-resolved datasets published for the community.

The same hybrid pattern (GPU-native structured far field, unstructured near-body) appears in [\[O2\]](#) and [\[O26\]](#).

Precision belongs in the error budget

Format	Bytes	Bandwidth-bound gain
FP64	8	1x
FP32	4	2x
TF32 / BF16	4 / 2	2x to 4x, tensor cores
FP8	1	8x, ML only today

- Every halving of bytes is a direct speed-up for a bandwidth-bound kernel.
- Tensor cores reward matrix-shaped work: dense blocks, high-order elements, batched small systems.

But precision is a modelling choice

- LAVA: single or mixed precision could give up to 2x more speed, but early trials showed "significant sensitivity of $C_{L,max}$ " to precision, so double precision remains their best practice.
- Explicit finite-difference turbulence tests: mixed precision works; pure half precision does not (Future Generation Computer Systems, 2025).
- Recommendation: mixed precision checked against a full-precision reference, per case class, reported like a mesh study.

Why this matters for data generation: flops per cell per step

Implicit steady RANS

- 80% or more of wall time in the linear solve.
- Few steps, expensive steps.

Explicit WMLES

- Cheap, deterministic step; cost driven by millions of small time steps.
- Maps naturally to GPUs and to online training (later).

Metric f : flops per cell per step

$$f_{\text{RANS}} \approx 40,000$$

$$f_{\text{WMLES}} \approx 2,800$$

A GPU-native transient solver is the cheapest known machine for producing training data. This is the hinge between claim 1 and claim 2.

Ashton, Brandstetter, Mishra, Fluid Intelligence, arXiv:2511.20455. Estimated ranges: 20k to 60k for implicit RANS, 1.8k to 3k for explicit WMLES; many production codes sit well above these.

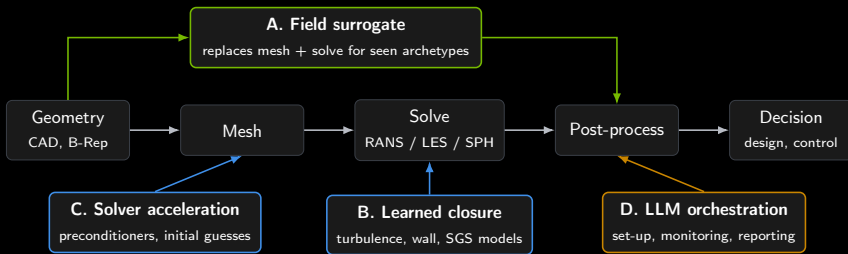
Weather was the proof point for learned surrogates of PDE systems



- Trained on 40 years of ERA5 reanalysis: the one place where a large, consistent, curated PDE dataset already existed.
- Forecasts in seconds on one GPU versus hours on a CPU supercomputer, at comparable or better skill for many variables.
- Direct relevance here: wind resource, wave climate and extreme-event statistics are downstream of exactly these models.
- The lesson is not the architecture. It is that the data existed.

Pathak et al. 2022; Bi et al. 2023; Lam et al. 2023; Price et al. 2024; ECMWF AIFS operational Feb 2025.

Where learning enters an engineering simulation workflow



This talk concentrates on A and D. B and C are large fields in their own right: see [\[O9 Castagna\]](#) and [\[O10 Farea\]](#) this afternoon for B-type work. The multiphysics version of this pipeline is [\[Keynote 2, Dubas\]](#) after lunch.

Architectures and inductive biases you will hear about today

Neural operators

FNO, GNO. Learn maps between function spaces; resolution-independent in principle.

[O8 Tan] [P7 Zhao]

Graph networks

Message passing on meshes and point clouds. Natural for unstructured CFD and particles.

[O9 Castagna] [P11 Liu]

Transformers

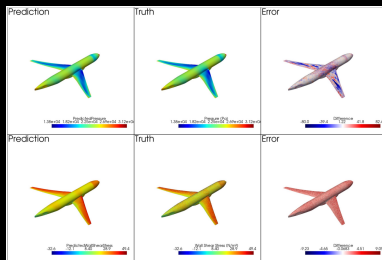
Attention over geometry tokens. Transolver, AB-UPT, GeoTransolver. Scale well with data.

[next slide]

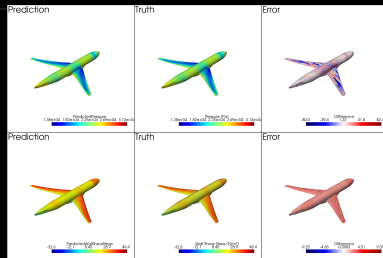
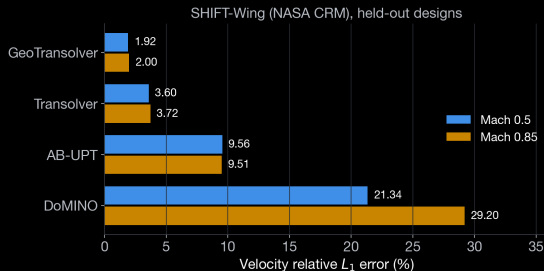
Physics-informed

PDE residual in the loss. Powerful for inverse problems and sparse data; hard at industrial scale.

[O10 Farea]



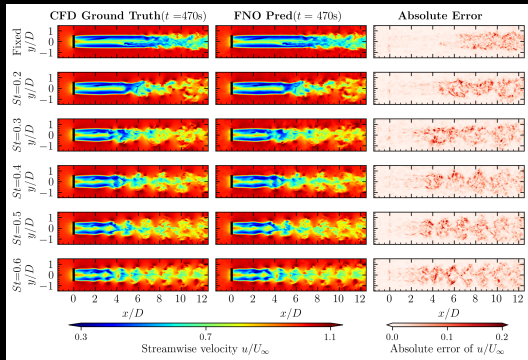
Evidence: transonic wing surrogates on held-out designs



- Same architecture family, same data: a 2x spread in error from inductive bias alone.
- Force coefficients are not discriminating (all $R^2 \approx 1$). Field errors on unseen designs are the honest metric.
- The Mach 0.85 case includes shocks. Errors barely move. Generalisation across regimes is possible with the right context.

GeoTransolver v2, arXiv:2512.20399, Tables 8 and 9. DoMINO trained separately per Mach number in this experiment.

Wind and wave examples: the same ideas, this community's problems



FOWT wake under platform surge at several Strouhal numbers: CFD, FNO prediction, error (arXiv:2604.23937).

Wind-farm wake operator (DTU, 2025)

Graph neural operator with a learned superposition principle. Trained on 10,000 simulated farms. Unseen test set: RMSE 0.353 m/s, MAPE 0.94%. Slight under-prediction of wake extent in dense layouts.

Wave-structure interaction

Graph neural operators for focused-wave loading [O8 Tan, Hartree]; neural-operator sloshing benchmark [P7 Zhao, Oxford]; the CCP-WSI AI hackathon on a buoy in focused waves [P12 Chen].

Each is a custom model on a custom dataset, which is where language models stood before 2018.

Labels have provenance: the same geometry, two different answers

DrivAerML

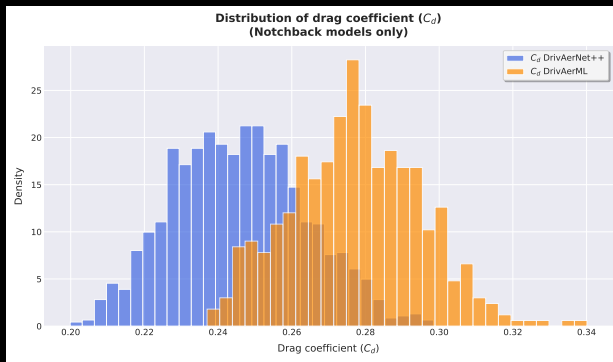
- 487 simulations, transient hybrid RANS-LES, 160M cells

DrivAerNet++

- 8,000 simulations, steady RANS, 23M cells

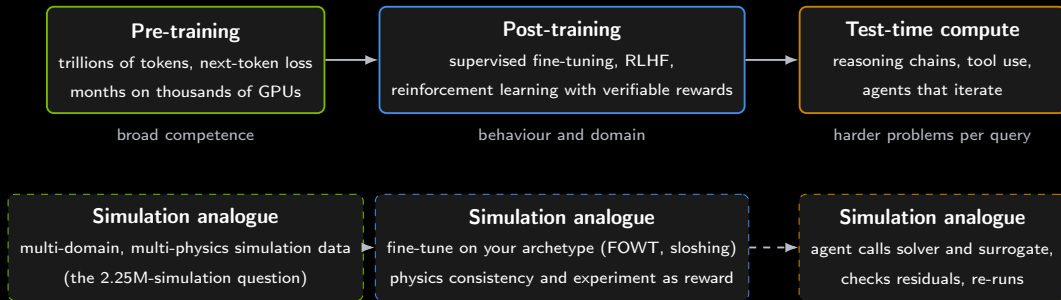
Consequence

Different solver choices produce different output distributions for the same design space. Pooling datasets blindly can make a model worse.



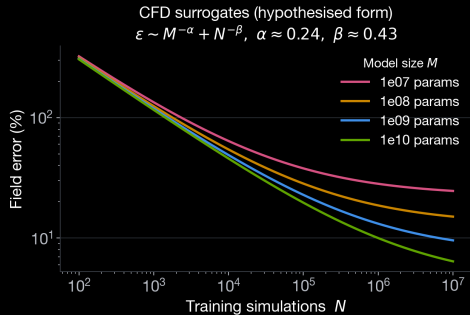
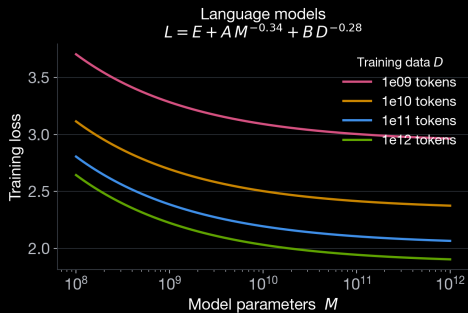
Ashton et al. 2024 (DrivAerML); Elrefaie et al. 2024 (DrivAerNet++). Analysis from arXiv:2511.20455. The same caution applies to wave tanks vs CFD vs field data.

How a large language model is actually built



Most of the cost sits on the left; most of the usefulness is unlocked in the middle and right. The same is likely true for simulation.

Scaling laws: the LLM result and the CFD hypothesis



- Left: measured. Loss falls as a power law in both parameters and data, and there is a compute-optimal balance between them (Hoffmann et al. 2022).
- Right: hypothesised form. The exponents are scenario parameters read from the Poseidon PDE foundation-model sweeps, not measurements on industrial 3D CFD. The data exponent β sets the whole cost curve and is the one to measure.

Left: Hoffmann et al. 2022 fit. Right: exponents from Herde et al. 2024 as used in arXiv:2511.20455; curves illustrative.

You cannot scrape the internet for CFD: the input space

The conditional distribution a simulation foundation model must learn

$$\mathbf{v}_{\text{CFD}} = [\mathbf{v}_{\text{geom}}, \mathbf{v}_{\text{prep}}, \mathbf{v}_{\text{BC}}, \mathbf{v}_{\text{mesh}}, \mathbf{v}_{\text{phys}}, \mathbf{v}_{\text{discr}}]$$

- $\mathbf{v}_{\text{geom}}$: geometry (B-Rep, NURBS, STL, moving bodies)
- $\mathbf{v}_{\text{prep}}$: clean-up and defeaturing decisions
- $\mathbf{v}_{\text{BC}}$: inflow, waves, free surface, mooring
- $\mathbf{v}_{\text{mesh}}$: topology and resolution, or particle spacing
- $\mathbf{v}_{\text{phys}}$: RANS, LES, SPH, two-phase, FSI
- $\mathbf{v}_{\text{discr}}$: schemes, time step, precision

For wave-structure interaction every one of these is live: a focused wave, a moving FOWT platform, a two-phase interface and a mooring model all sit in the input vector. Text tokens have no equivalent.

How much data, and what would it cost?

$$9 \times 100 \times 50 \times 50 = 2.25 \text{ M}$$

Domains

aero, auto, wind, marine...

Archetypes

sedan, FOWT, monopile...

Parameters

Re, AoA, wave height...

Values per parameter

samples per axis

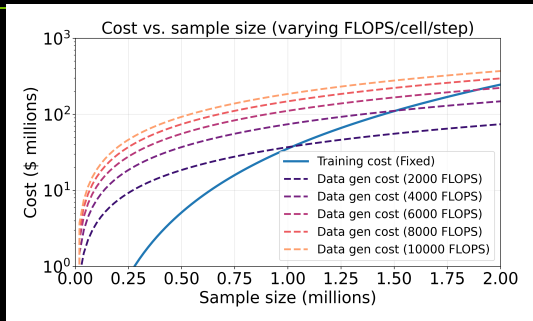
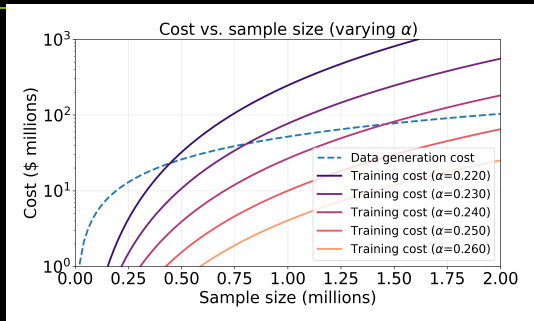
simulations

	LF (steady RANS)	HF (LES, averaged)	HF-T (LES, transient)
Training samples N	3.25×10^5	10^6	2×10^5
Cells / steps	10^8 / 2,000	5×10^8 / 200,000	5×10^8 / 200,000
GPU hours	1.8×10^5	6.5×10^6	1.3×10^6
Data-generation cost	\$1.4M	\$51.8M	\$10.3M
Storage (yearly cost)	650 TB (\$78k)	2 PB (\$240k)	40 PB (\$4.8M)

arXiv:2511.20455, Table 3. FP32 on NVIDIA GB200 at \$8 per GPU hour; 15% sustained for explicit WMLES, 5% for implicit RANS. Transient

runs store 100 states each, hence the storage jump; they need fewer samples because each trajectory carries many snapshots.

Cost sensitivity and the regime flip



- For a "Large" model (2×10^5 transient samples) data generation is about 96% of the budget: \$10.3M against \$0.4M. Faster solvers are the lever.
- At 2×10^6 samples the regime flips: training reaches about \$246M against \$104M for data. Model efficiency becomes the lever.
- Physics parameters (cells, steps, flops per cell) shift the data-generation line but not its slope. The exponents set the slope.

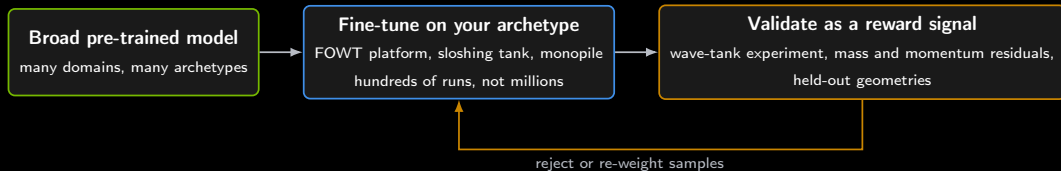
Pre-training candidates exist. What have they shown?

	Demonstrated	Not yet demonstrated
Poseidon (ETH, 2024)	Pre-trained on PDE families (Euler, Navier-Stokes); transfers to 15 downstream tasks with far fewer samples than training from scratch.	Industrial 3D geometry, high Reynolds number, unstructured meshes, free surfaces.
Walrus (Polymathic, Nov 2025)	Continuum-dynamics pre-training across 19 scenarios, mixed 2D and 3D, with useful transfer.	Engineering boundary conditions, moving bodies, mooring, validation against experiment.
Domain models (2024 to 2026)	On 48 held-out DrivAerML designs, GeoTransolver reaches drag $R^2 = 0.996$ and 2.9% surface-pressure error.	Cross-domain transfer: car to turbine to platform.

The gap is not architecture. It is the input space on the previous slide and the data on the one before it.

Herde et al., arXiv:2405.19101; Polymathic AI, arXiv:2511.15684; GeoTransolver v2, arXiv:2512.20399, Table 1.

Post-training is where this community's data matters most



What you already have

Benchmark experiments [O16 sloshing, Plymouth, tomorrow], reference FOWT datasets [O27 DTU 15 MW tandem, tomorrow], the CCP-WSI hackathon cases [P12], and open high-fidelity data from the GPU codes in Session 1.

What the LLM analogy suggests

Supervised fine-tuning needs curated, consistent labels. Reward-style training needs a verifier: an experiment or a conservation check is a far better verifier than a human preference.

Future algorithms: co-designing solvers for learning

Transient high-fidelity data

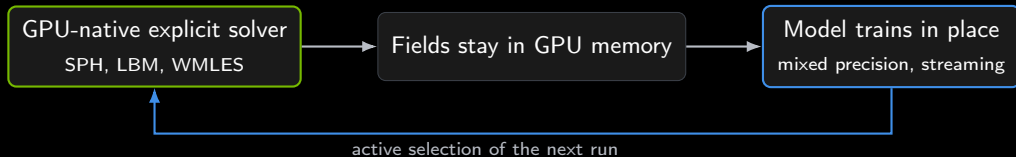
Trajectories carry more information per simulation than averages. The HF-T column was five times cheaper than HF for the same target.

Online, in-situ training

Train while the solver runs, on the GPU, with no intermediate storage. The challenge is matching solver time-scales to back-propagation.

Compression

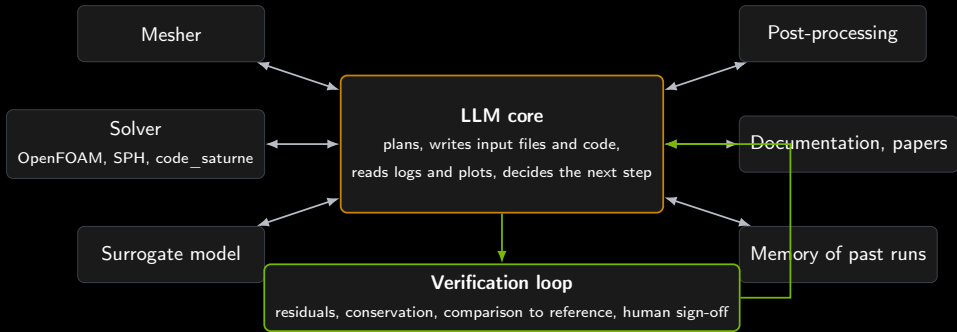
Petabytes per dataset outrun HPC storage. POD and learned latent spaces instead of subsampling.



SPH and lattice Boltzmann are unusually well placed here: explicit, local, and already GPU-resident.

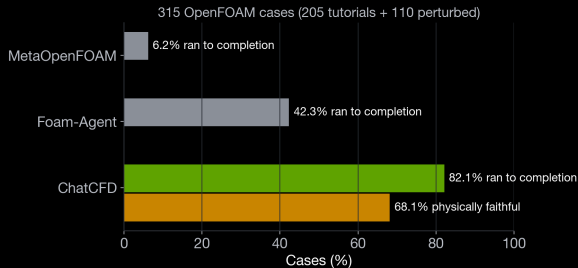
[O3 Rogers] [O14 Rogers] [O15 Tsigginos]

What an agent is, in one picture



The tools are the ones this community writes. The agent adds a loop around them. Whether the loop converges depends on tool speed and on what the verifier can catch.

Evidence: agents running OpenFOAM end to end

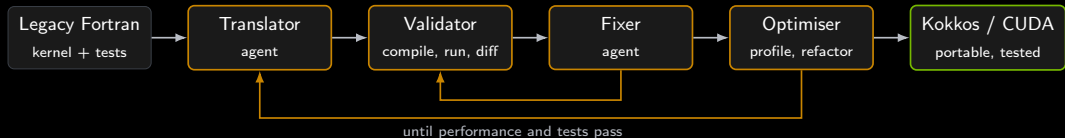


- Execution success and physical fidelity are different things. A case that runs is not a case that is right.
- The physics interpreter summarised the intended set-up correctly 97% of the time; enforcing dozens of coupled constraints in executable files is the hard part.
- Structured knowledge (solver templates, error locator) mattered more than the base model.
- Cost per case around \$0.20 and 190k tokens. Cheap enough to use routinely; not yet trustworthy enough to use unsupervised.

Fan et al., ChatCFD, Advanced Intelligent Discovery (2026) doi:10.1002/aidi.202500174; Foam-Agent 2.0 arXiv:2509.18178; MetaOpenFOAM.

Benchmark excludes cases needing complex mesh generation.

Agents that port codes: a direct offer to Session 1



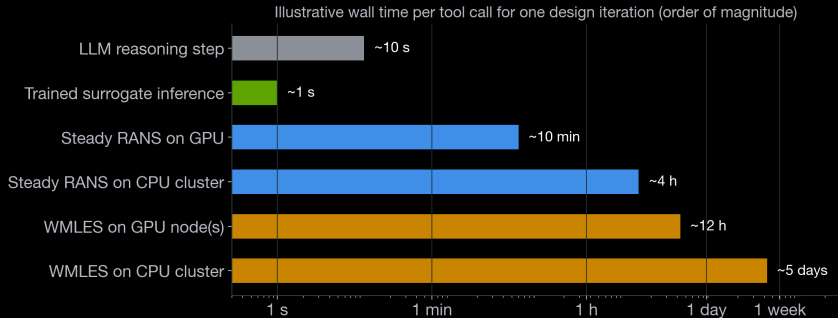
Published results

Fortran to Kokkos with translator, validator, fixer and optimiser agents on benchmark kernels for a few dollars per kernel (arXiv:2509.12443). C++ to CUDA via de-optimize then re-optimize (arXiv:2606.06063). Differential testing of LLM-translated HPC codes (Kaizen, arXiv:2607.04058).

Honest limits

Kernels, not whole codes. Correctness needs a real test suite; the agent cannot invent one. Performance tuning still needs someone who knows the memory hierarchy. The porting work in [O3 DualSPHysics] [O4 CHAPSim] [O5 gpuFOAM] is exactly the shape these tools help with.

Slow tools limit the agent's loop



- An agent that iterates ten times a day against a CPU WMLES gets nowhere. Against a GPU solver plus a validated surrogate it can explore a design space overnight.
- This is the strongest argument for GPU-native solvers that is not about any single simulation.

Order-of-magnitude, illustrative values for a mid-size external-flow case; not measurements.

Open problems where the CCPs are well placed

Data and benchmarks

- Open, consistent, high-fidelity datasets for wave-structure interaction and floating wind, with held-out geometries. DrivAerML, WindsorML and AhmedML are templates.
- Benchmarks that report field error on unseen designs, not only forces. [\[P12 hackathon\]](#) is the right instinct.

Solvers

- GPU-native, open, explicit codes that expose fields in memory for online training. [\[O3\]](#) [\[O5\]](#) [\[O15\]](#)

Models

- Measure the data exponent β for one archetype. It is the most valuable single number in this whole cost analysis.
- Uncertainty: a surrogate without an error bar cannot be a tool in a loop.

Agents

- Physical-fidelity benchmarks for agents on the community's own tutorial suites (SPH, `code_saturne`, OpenFOAM).
- Verifiers built from experiments: the wave tanks in this room are reward functions.

Conclusions

1. **Accelerated computing** has already changed what is affordable. Throughput and energy gains of an order of magnitude are measured, not projected, when the algorithm fits the hardware.
2. **Surrogates** are accurate enough to be useful on seen archetypes. Their limits are set by data provenance and coverage, not by architecture.
3. **The LLM playbook** translates: expensive pre-training on broad data, cheap and decisive post-training on your own archetype, verifiers instead of preferences. The data exponent decides the economics.
4. **Agents** work today for set-up, monitoring and porting, and fail quietly on physics. Fast GPU solvers, validated surrogates and experimental verifiers are what make them trustworthy.

The scarce ingredients

The communities in this room own the solvers, the experiments and the benchmarks. Those are the scarce ingredients.

Thank you

Fluid Intelligence: A Forward Look on AI Foundation Models in CFD

<https://arxiv.org/abs/2511.20455>

With thanks to co-authors Prof. Siddhartha Mishra (ETH) and Prof. Johannes Brandstetter (JKU Linz), and to the LAVA, ExaWind, DTU and ChatCFD teams whose results are reproduced here.

Open datasets: <https://caemldatasets.org>

Backup: asymptotic complexity of the three surrogate regimes

Model	Error	Model size	Dataset size	Total compute
S_r^θ (LF)	$\epsilon_r \sim N_r^{-\beta}$	$M_r \sim N_r^{\beta/\alpha}$	N_r	$nf_e \left(N_r^{1+\beta/\alpha} + N_r^{1+3\beta/\gamma} \right) + f_r N_r^{1+(3+\kappa)\beta/\gamma}$
S_ℓ^θ (HF)	$\epsilon_\ell \sim N_\ell^{-\beta}$	$M_\ell \sim N_\ell^{\beta/\alpha}$	N_ℓ	$nf_e \left(N_\ell^{1+\beta/\alpha} + N_\ell^{1+3\beta/\gamma} \right) + f_\ell N_\ell^{1+4\beta/\gamma}$
S_t^θ (HF-T)	$\epsilon_\ell \sim (\delta_K \tilde{N}_\ell)^{-\beta}$	$M_t \sim \tilde{N}_\ell^{\beta/\alpha}$	$\tilde{N}_\ell$	$nf_e K \left(\tilde{N}_\ell^{1+\beta/\alpha} + \tilde{N}_\ell^{1+3\beta/\gamma} \right) + f_\ell \tilde{N}_\ell^{1+4\beta/\gamma}$

- α model exponent (≈ 0.24), β data exponent (≈ 0.43), γ numerical convergence rate (≈ 1), κ linear-solver cost exponent.
- n epochs, f_e flops per sample per epoch, f_r , f_ℓ flops per cell per step for RANS and LES.
- Transient case: K effective copies per trajectory (50) and compression ratio δ_K (5), so 2×10^5 transient runs stand in for 10^6 averaged ones.
- Since $3\alpha < \gamma$, the processor dominates the encoder in training cost.

Backup: model training costs alongside data generation

	LF	HF	HF-T		Large	XLarge	XXLarge
Training samples N	3.25×10^5	10^6	2×10^5	Transient samples N	2×10^5	10^6	2×10^6
Data-generation cost	\$1.4M	\$51.8M	\$10.3M	Data-generation cost	\$10.3M	\$51.8M	\$103.7M
Model size M	7.5B	56B	3.2B	Storage	40 PB	200 PB	400 PB
Training cost	\$31k	\$710k	\$397k	Model size M	3.2B	56B	195B
Training share of total	2%	1%	4%	Training cost	\$0.4M	\$35.5M	\$245.7M

- Below about 10^6 samples the solver is the bottleneck, not the model.
- At 2×10^6 transient samples, close to the independent 2.25×10^6 coverage estimate, training overtakes data generation: the regime flip.

Backup: references

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